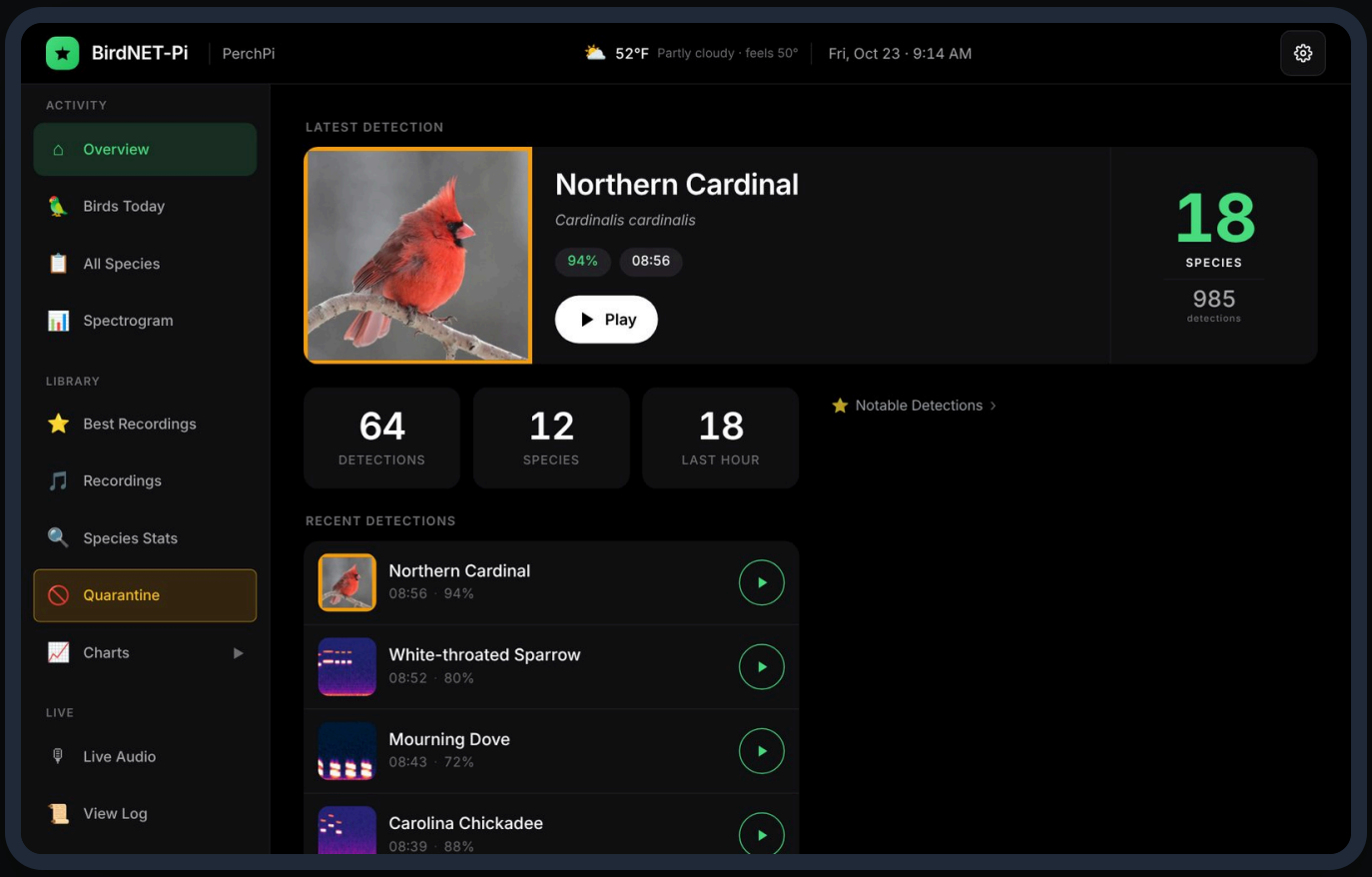
PerchPi

Your backyard bird station

Setup Guide & User Manual · PerchPi v1 for Raspberry Pi 5



PerchPi listens around the clock, names the birds it hears, keeps their songs, and shows it all on any phone, computer, or touchscreen in your home.

Help: **perchpi@shadyacres.my**
github.com/churchja/perchpi-release

Screens show example data.
October 2026

START HERE

Quick start

The whole setup on one page. Each step has its own page in Part 1 with pictures.

1 Get the parts

Raspberry Pi 5 (8 GB), Active Cooler, 27W power supply, Samsung 64 GB microSD, **Clippy EM272** microphone, **UGREEN US205** adapter.

2 Connect

Microphone into the **red** jack of the adapter, adapter into a USB port. Cooler on the Pi.

3 Install Raspberry Pi Imager

Version 2.0 or newer, from raspberrypi.com/software.

4 Point Imager at PerchPi

App Options → Content Repository → Edit → Use custom URL, paste the address below, **Apply & Restart**.

5 Write the card

Raspberry Pi 5 → PerchPi v1 → your card. Fill in hostname, username and password, Wi-Fi, and time zone. Write.

6 Start it

Card in, power on, wait about 5 minutes. It restarts once by itself.

7 Open it

`http://perchpi.local/` on the same Wi-Fi. Web login `birdnet` / `birdnet`.

8 Two settings

Change the web password and set your location (gear icon, top right).

```
https://github.com/churchja/perchpi-release/releases/latest/download/perchpi-os-list.json
```

⚠ Use the Clippy EM272 microphone and the UGREEN US205 adapter. With another microphone, detection may be severely degraded or nearly useless. With another adapter, PerchPi v1 may record nothing at all.

INSIDE THIS GUIDE

Contents

This guide online
Scan for the latest version of the Setup Guide.

Raspberry Pi Imager
Free download for Mac, Windows, Linux.

Part 1 · Set up

1 · What you need
2 · Put it together
3 · Install Imager
4 · Point Imager at PerchPi
5 · Write the card
6 · First start
7 · Open PerchPi
8 · Three settings
9 · Test it
10 · Place the microphone

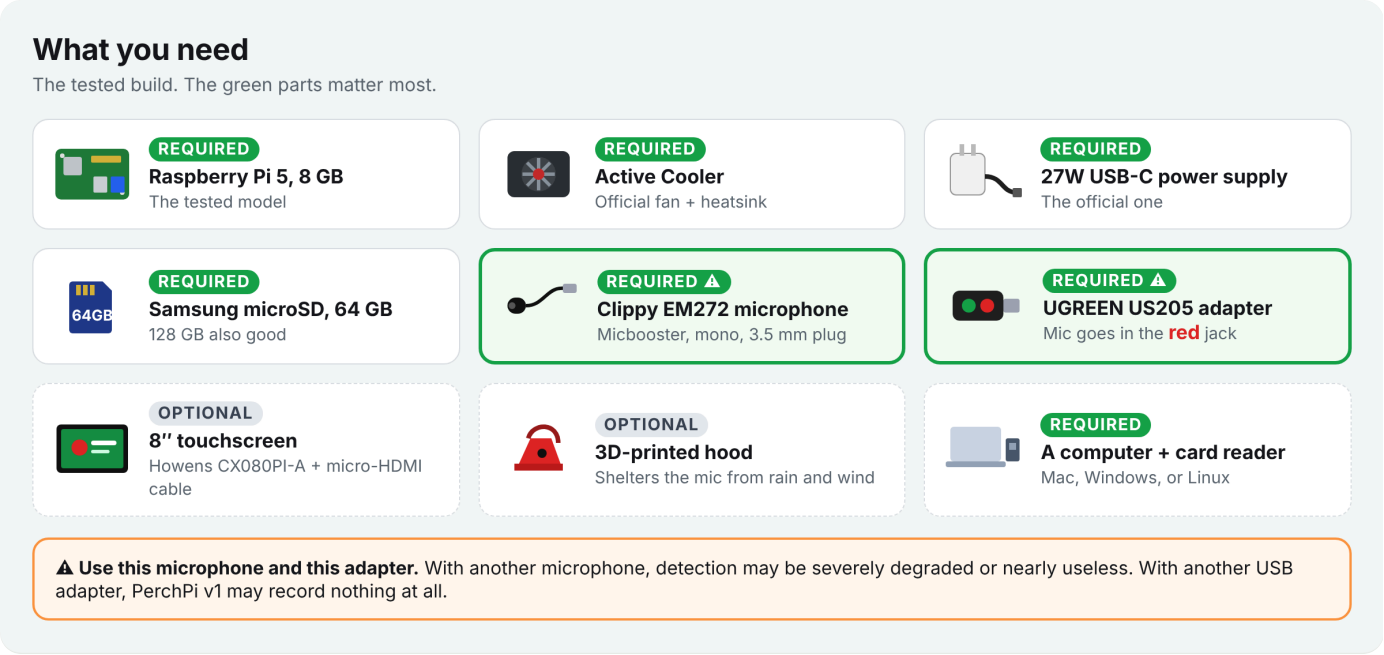
Part 2 · Use PerchPi

The Overview page
The menu
Birds and recordings
Reviewing Quarantine
Touchscreen and phone
Settings you might want
Storage
Turning it off and on
Backing up and updating
Troubleshooting · Getting help

PART 1 · SET UP

1 What you need

These are the exact parts PerchPi is built and tested with. Buy these.

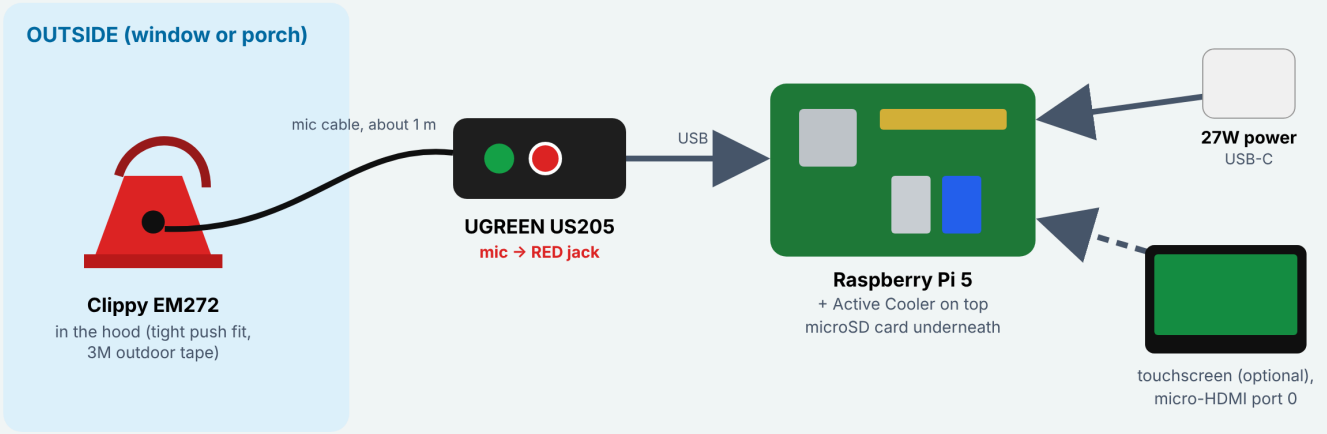


- **Raspberry Pi 5, 8 GB**: raspberrypi.com/products/raspberry-pi-5
- **Raspberry Pi Active Cooler**: PerchPi runs day and night; the cooler stops it slowing down when warm.
- **Raspberry Pi 27W USB-C Power Supply**, the official one.
- **Samsung microSD, 64 GB** (128 GB also good). Every recording and the bird database live on it.
- **Clippy EM272 Mono Microphone**, Micbooster (micbooster.com): the plain mono version with the 3.5 mm plug, not XLR or "Smart".
- **UGREEN USB 3.5 mm External Sound Adapter, model US205** (code 30724): green jack for headphones, red jack for the microphone.
- **Optional:** the 8″ Howens CX080PI-A touchscreen (needs a micro-HDMI-to-HDMI cable), and the 3D-printed hood (files: github.com/churchja/perchpi-release/tree/main/hardware/em272-hood).

2 Put it together

How it connects

Only the microphone goes outside. Everything else stays indoors and dry.



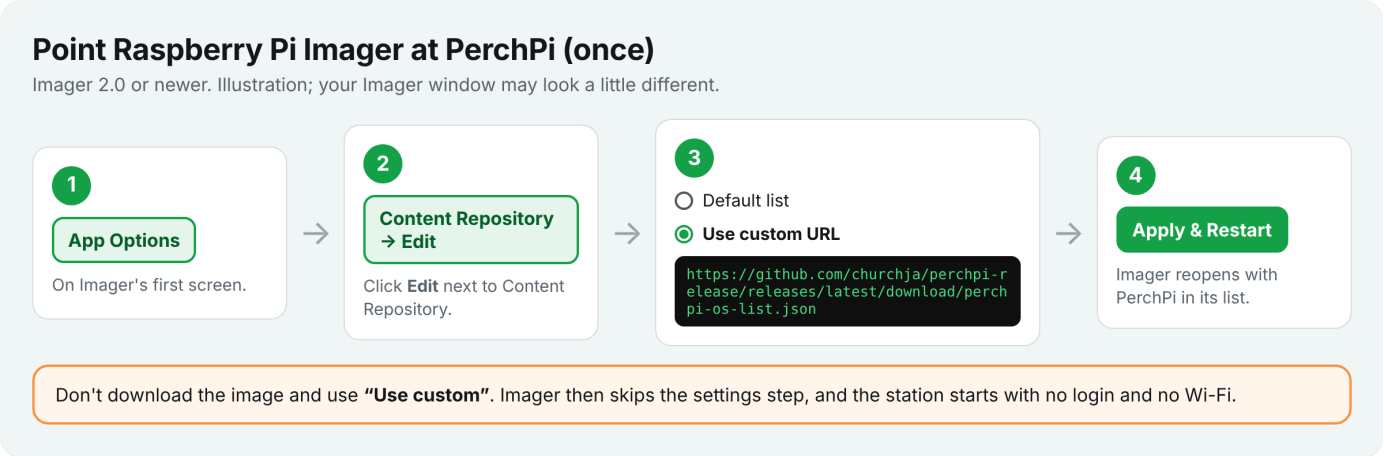
1. **Fit the cooler.** Peel the film off its thermal pads, line its two spring-loaded pins up with the holes in the Pi, and press until they click. Plug its small cable into the 4-pin socket labelled **FAN**.
2. **Connect the microphone** to the **red** jack of the UGREEN adapter (not the green one). Plug the adapter into any USB port on the Pi.
3. **Touchscreen (optional):** the Pi's micro-HDMI socket next to the power socket (HDMI 0), and its touch cable to a USB port.
4. **Leave the power off.** The memory card comes first.

3 Install Raspberry Pi Imager

Imager is the free program that writes PerchPi onto the memory card. Download it from **raspberrypi.com/software** and install it like any other program. You need **version 2.0 or newer**; if you already have Imager, update it.

4 Point Imager at PerchPi

PerchPi isn't in Imager's built-in list, so you tell Imager where to find it. You only do this once.



The address to paste:

```
https://github.com/churchja/perchpi-release/releases/latest/download/perchpi-os-list.json
```

Shortcut for Mac users comfortable with Terminal. Open the Terminal app, paste this line, and press Return. Imager opens with PerchPi's list.

```
osascript -e 'quit app "Raspberry Pi Imager"'; sleep 2; open -a "Raspberry Pi Imager" --args --repo https://github.com/churchja/perchpi-release/releases/latest/download/perchpi-os-list.json
```

Linux: `rpi-imager --repo` followed by the address. Copying from a PDF can split the line, so copy it from the online Setup Guide (QR code on the Contents page). If a shortcut doesn't work, use the menus.

To go back to Raspberry Pi's normal list later, open **App Options → Content Repository → Edit** again and choose the default.

5 Write the memory card

Choose, customise, write

Illustration of the choices to make in Imager.

1 DEVICE
Raspberry Pi 5

2 OPERATING SYSTEM
PerchPi v1
Bird-detection station for Raspberry Pi 5

3 STORAGE
Your microSD card
Check the size: it will be erased

4 Customisation: fill in every line

Hostname	`perchpi` One word: letters, numbers, hyphens. You'll type it in your browser.
Username / password	your own (for the Pi itself) Write it down. The web page uses a **different** login.
Wi-Fi + country	your network name, its password, your country
Time zone	where the station is
SSH	optional: leave off if unsure

5 **Write**. Your computer may ask for **its own** password. Writing and checking takes 10 to 30 minutes.

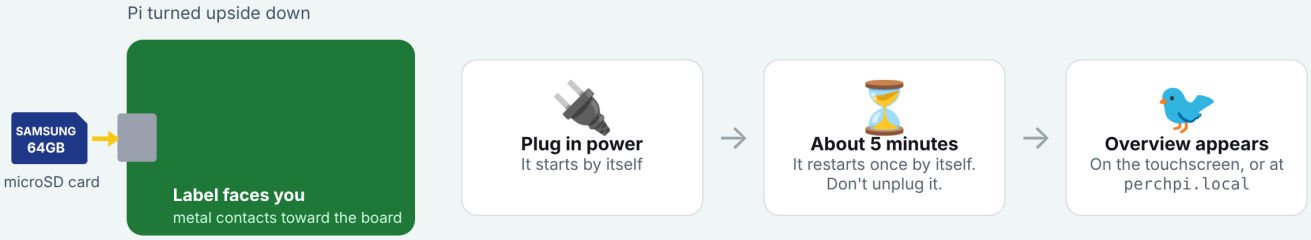
Check the storage choice. Everything on the chosen card or drive is erased.

Imager downloads PerchPi (about 2 GB), writes it, and checks it. Allow 10 to 30 minutes. When it says it's finished, take the card out.

6 First start

First start

Turn the Pi over. Slide the card into the slot until it stops, label facing you.



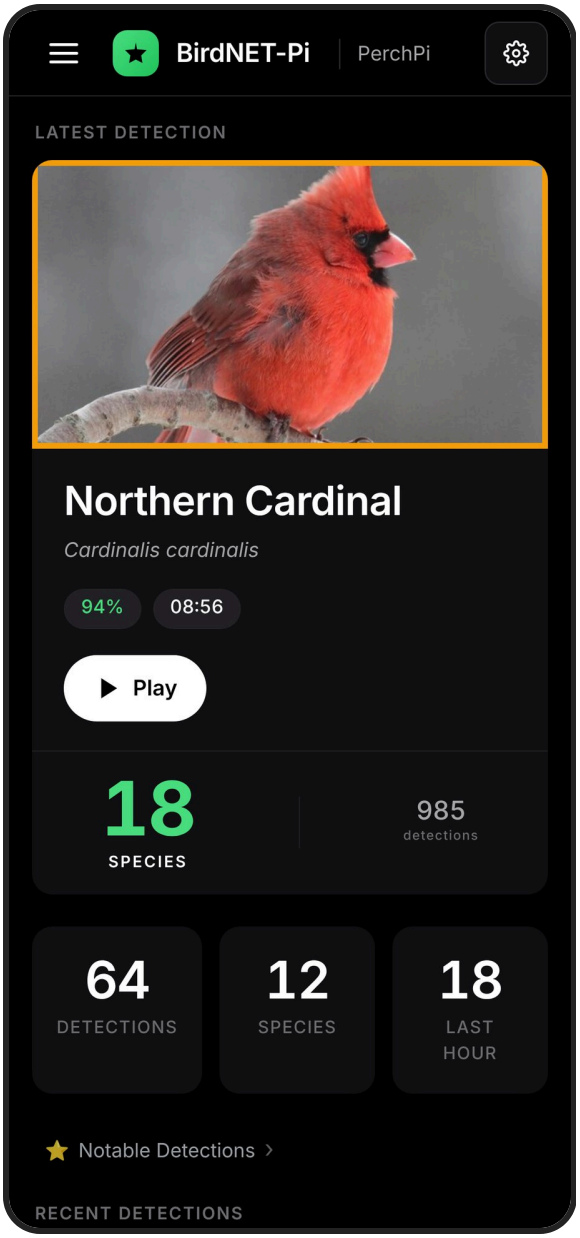
1. Put the card in the slot on the underside of the Pi.
2. Check the microphone adapter is plugged in.
3. Plug in the power. The Pi starts by itself; there's no need to press its small power button.
4. **Wait about 5 minutes.** On its first start, the Pi enlarges its storage and restarts once by itself. Don't unplug it.

7 Open PerchPi in your browser

On a phone, tablet, or computer **on the same Wi-Fi**, open a browser and go to `http://perchpi.local/`, using the hostname you chose (`backyard` → `http://backyard.local/`).

The page title says **BirdNET-Pi**. That's normal: PerchPi is built on BirdNET-Pi.

> **Address won't open?** Wait a few more minutes, and check you're on the same Wi-Fi. Some Android phones and older Windows computers can't open `.local` addresses: find the Pi's number address in your router's device list and type that instead, e.g. `http://192.168.1.57/`.



PerchPi on a phone (example data).

Two different logins

The one you typed in Imager is not the web page login.

🖥️ The Pi itself

Set by **you** in Raspberry Pi Imager

Only for advanced maintenance (SSH). Most people never use it.

🌐 The PerchPi web page

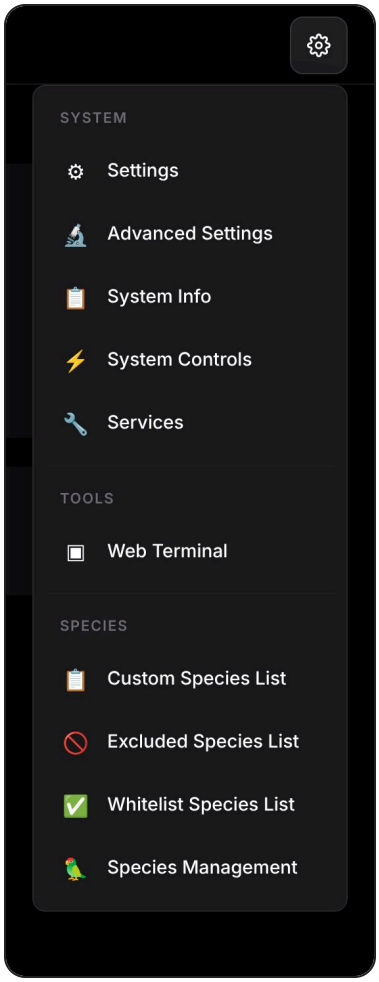
Username **`birdnet`**

Password **`birdnet`** until you change it

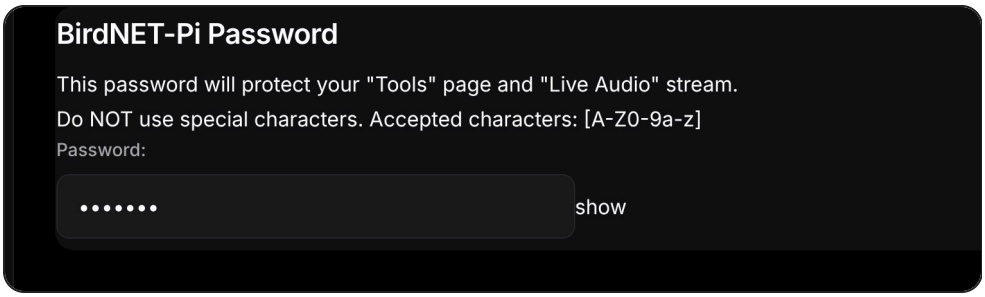
Change it: gear → Advanced Settings → BirdNET-Pi Password

8 Three settings to change right away

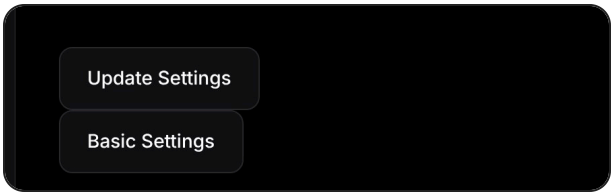
The **gear icon** at the top right opens the settings menu. On the settings pages, scroll down to the section, then click **Update Settings** at the bottom of the page.



The gear menu.



Advanced Settings → BirdNET-Pi Password.



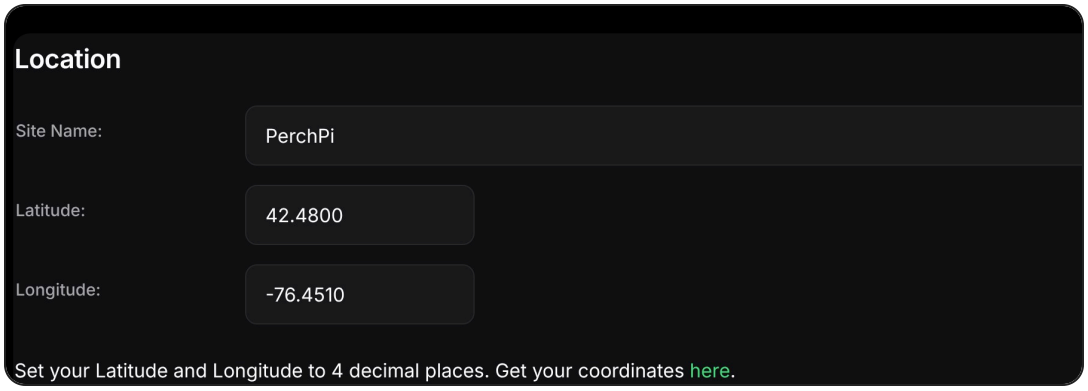
Every settings page saves with **Update Settings** at the bottom.

1 · Change the web password (important)

gear → Advanced Settings, scroll to **BirdNET-Pi Password**. Delete everything in the box, type a new password using **only letters and numbers**, and click **Update Settings**. Your browser asks you to log in again: `birdnet` and your new password. Write it down.

8 Three settings (continued)

2 · Set your location



Settings → Location (shown with the default location).

Open Google Maps on a computer and right-click your house. The first line shows two numbers, like `40.71278, -74.00601`. In **gear → Settings → Location**, type the first number into **Latitude** and the second into **Longitude**. Keep the minus sign; at most **4 digits after the point** (`40.7128`, `-74.0060`). Click **Update Settings**.

> Until you do this, PerchPi uses a location in New York State, so its bird suggestions and weather are wrong.

3 · Check the time zone

gear → Settings → Time and Date. It should already match what you chose in Imager.

9 Test it

On your phone, play a recording of a bird common where you live at this time of year (in most of the US, a Northern Cardinal) about an arm's length from the microphone, for 20 to 30 seconds. Within a minute or two it appears on the **Overview** and under **Birds Today**. Tap it to see the bird and hear what PerchPi recorded.

> **Not on the Overview?** Look in **Quarantine**. PerchPi holds birds there that are unlikely for your location and season.

10 Place the microphone

Where the microphone sits matters more than anything else.

- **Only the microphone goes outside.** Keep the Pi, power supply, and UGREEN adapter indoors and dry.
- **Outside is best**: an open or cracked window, a porch, or under an eave. The developer's microphone sits in a second-floor window and hears birds 25 to 35 feet away.
- **Shelter it from rain and wind** with the printed hood. Push the microphone in (it's a tight fit), then stick the hood up with **3M double-sided outdoor mounting tape**.
- **Away from noise:** air conditioners, pumps, wind chimes, busy roads, and the Pi's fan.
- The microphone cable is about **1 m (3 ft)**. A USB extension between adapter and Pi adds reach.

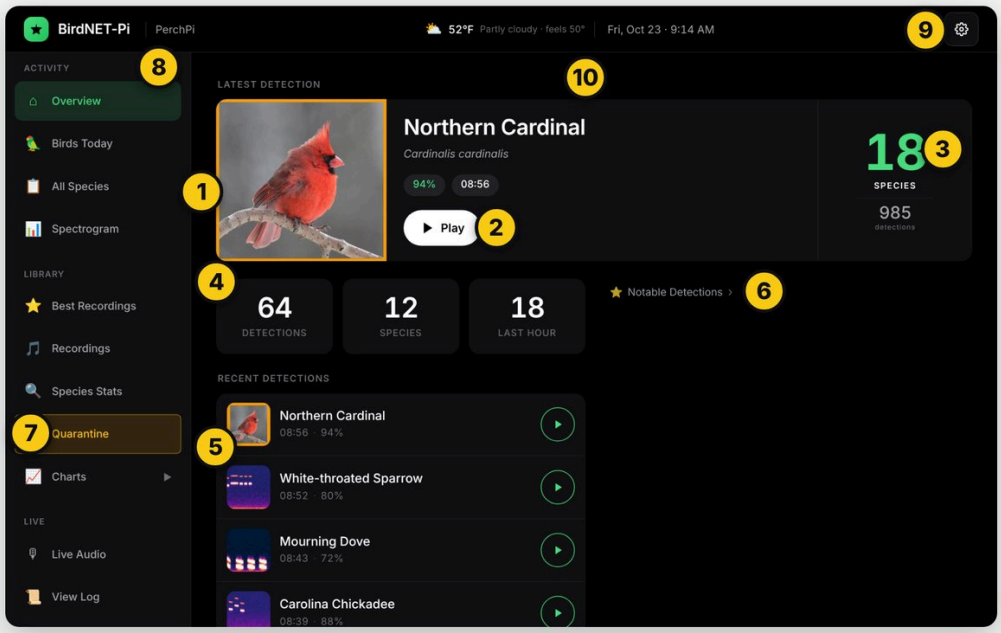
> **Print your own hood:** PETG, 0.2 mm layers, 2 walls, 15% infill, no supports, flat base down. About 50 minutes and 17 g. Files: github.com/churchja/perchpi-release/tree/main/hardware/em272-hood

PART 2 · USE PERCHPI

The Overview page

The Overview page

What you see when you open PerchPi (example data).



1. **Latest bird** — Its photo, name, how sure PerchPi is (94%), and when it was heard.
2. **Play** — Hear the recording PerchPi made.
3. **All-time totals** — Every species and detection so far.
4. **Today** — Detections, species, and the last hour.
5. **Recent birds** — Newest first. Tap ▶ to listen.
6. **Notable** — Rare or newly arrived birds.
7. **Quarantine** — Turns orange when there are unlikely birds to review.
8. **Menu** — Every page. On a phone, tap ☰.
9. **Gear icon** — Settings, password, shutdown.
10. **Weather and time** — For the location you set.

How PerchPi works, in one minute

- The microphone records continuously, in 15-second pieces. Each piece is checked by two bird-identification programs: **BirdNET** (Cornell Lab of Ornithology) and Google's **Perch**.
- A confident identification becomes a **detection**: species, time, confidence score, and the audio with a picture of the sound (a **spectrogram**).
- PerchPi knows which birds are likely where you live, week by week. A very unlikely bird goes to **Quarantine** for you to review.
- **Your recordings stay on the Pi.** It goes online for bird photos (Wikipedia) and weather, and shares detections only if you turn on BirdWeather or notifications.

The menu

Activity

- **Overview**: the home page.
- **Birds Today**: every species heard today, with counts.
- **All Species**: everything the station has ever heard.
- **Spectrogram**: a live picture of what the microphone hears.

Library

- **Best Recordings**: the clearest recording of each species.
- **Recordings**: every saved recording, by date and species.
- **Species Stats**: counts, first and last heard, activity.
- **Quarantine**: unlikely birds to review.

Charts

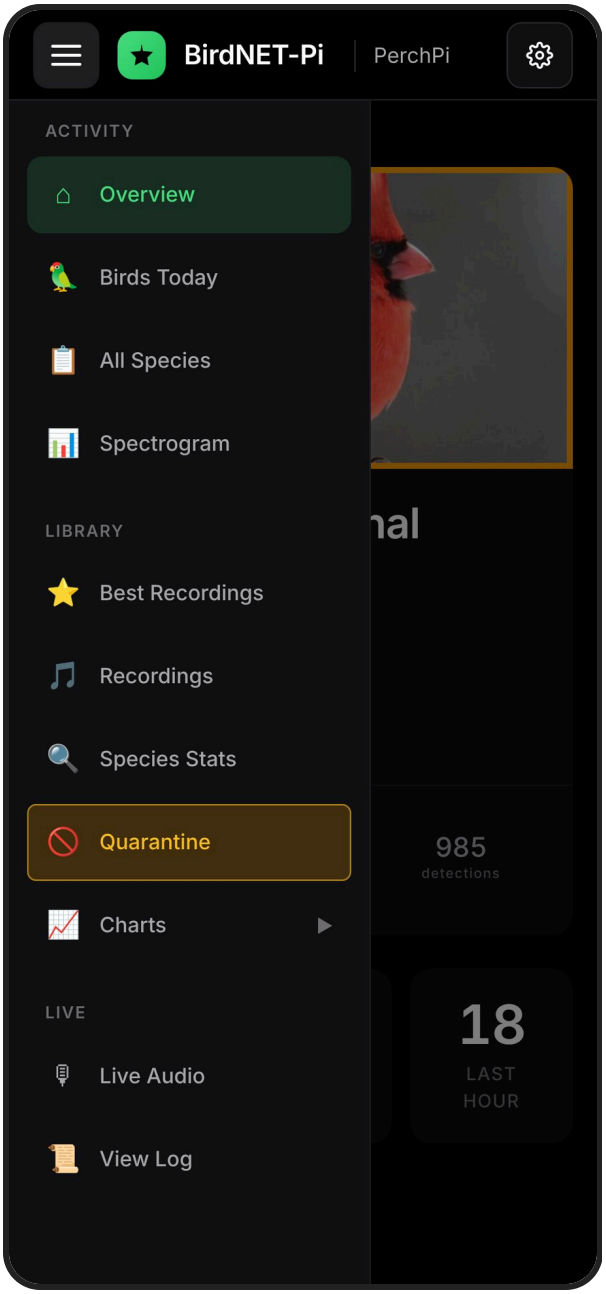
- **Daily Charts**, **Weekly Report**, **Detection Patterns**.

Live

- **Live Audio**: listen live (about 6 seconds behind). The quickest microphone check.
- **View Log**: the technical log.

Gear menu (top right)

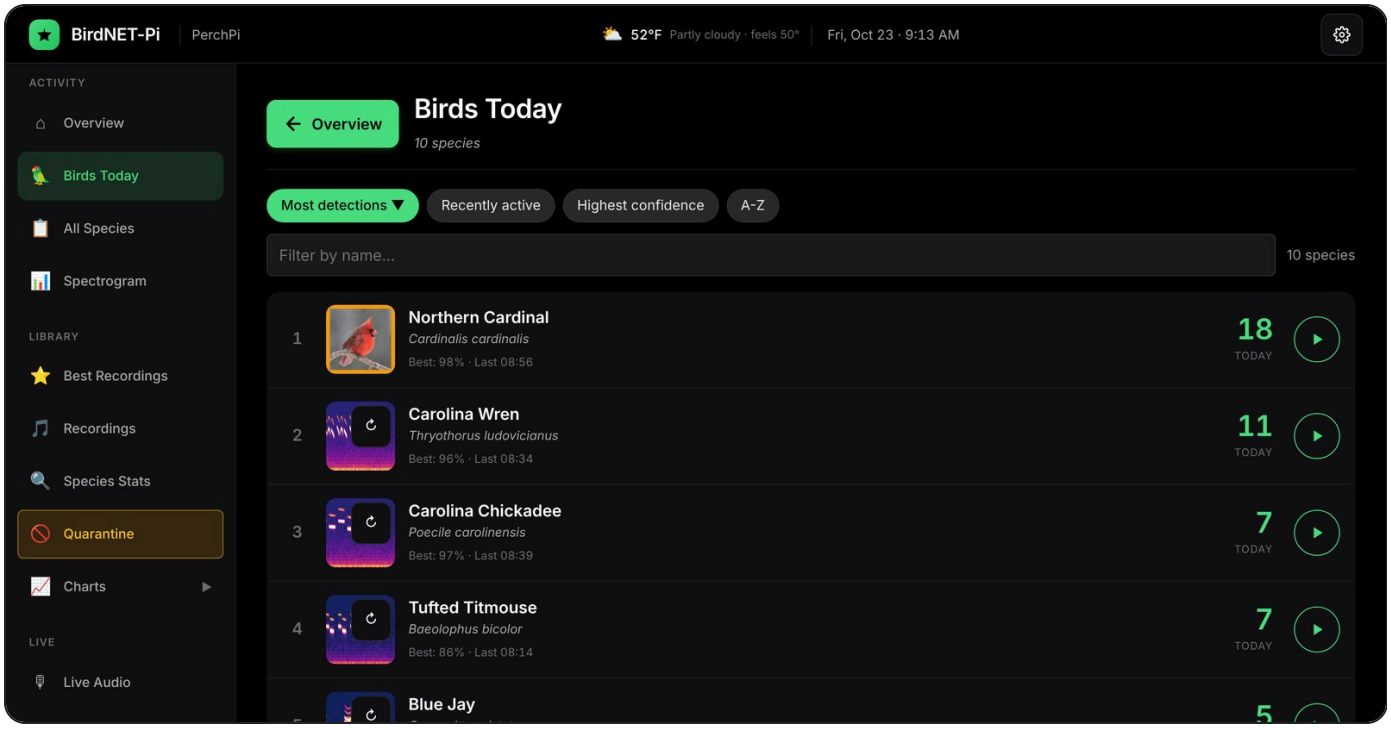
- **Settings**, **Advanced Settings**, **System Info**, **System Controls**, **Services**.
- **Web Terminal** is turned off in v1: ignore it.
- Species lists: **Custom**, **Excluded**, **Whitelist**, **Species Management**.



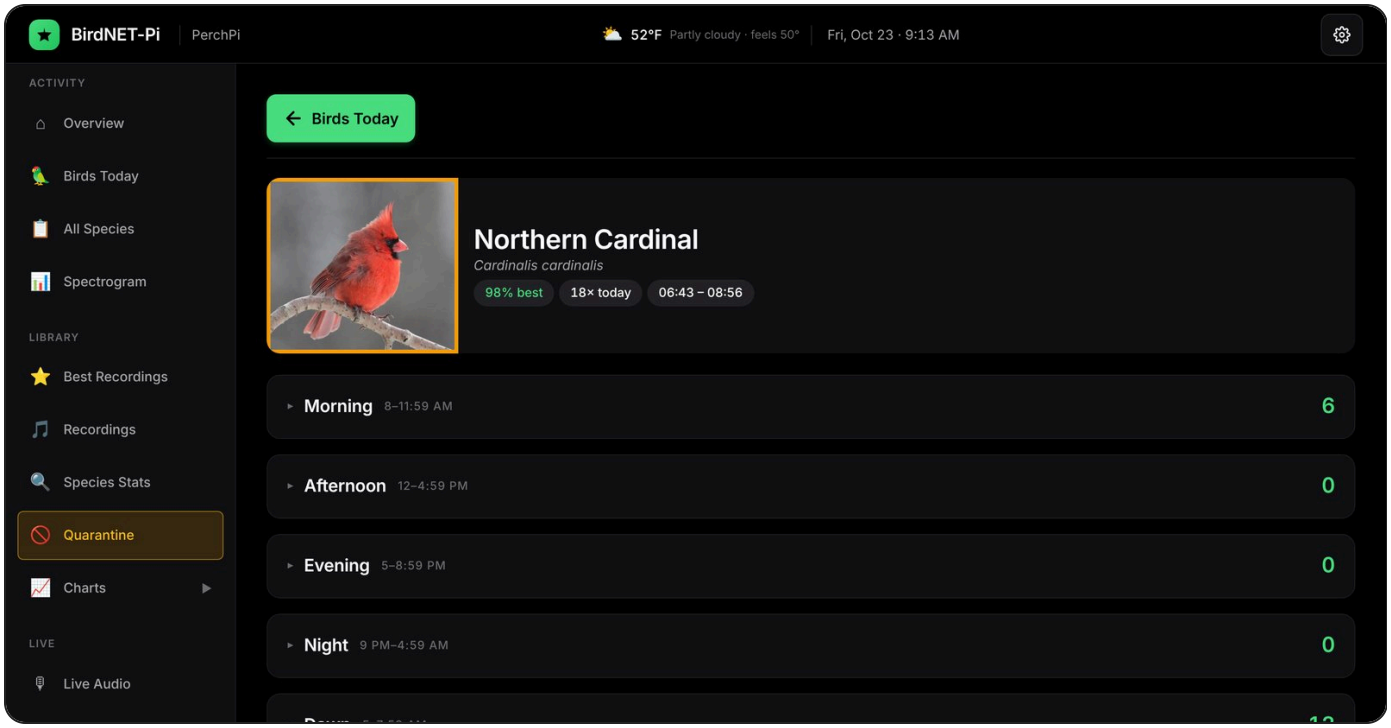
On a phone, tap ☰ (top left) to open the menu.

Login: most pages ask for the web login the first time you open them (`birdnet` and your web password). Your browser remembers it.

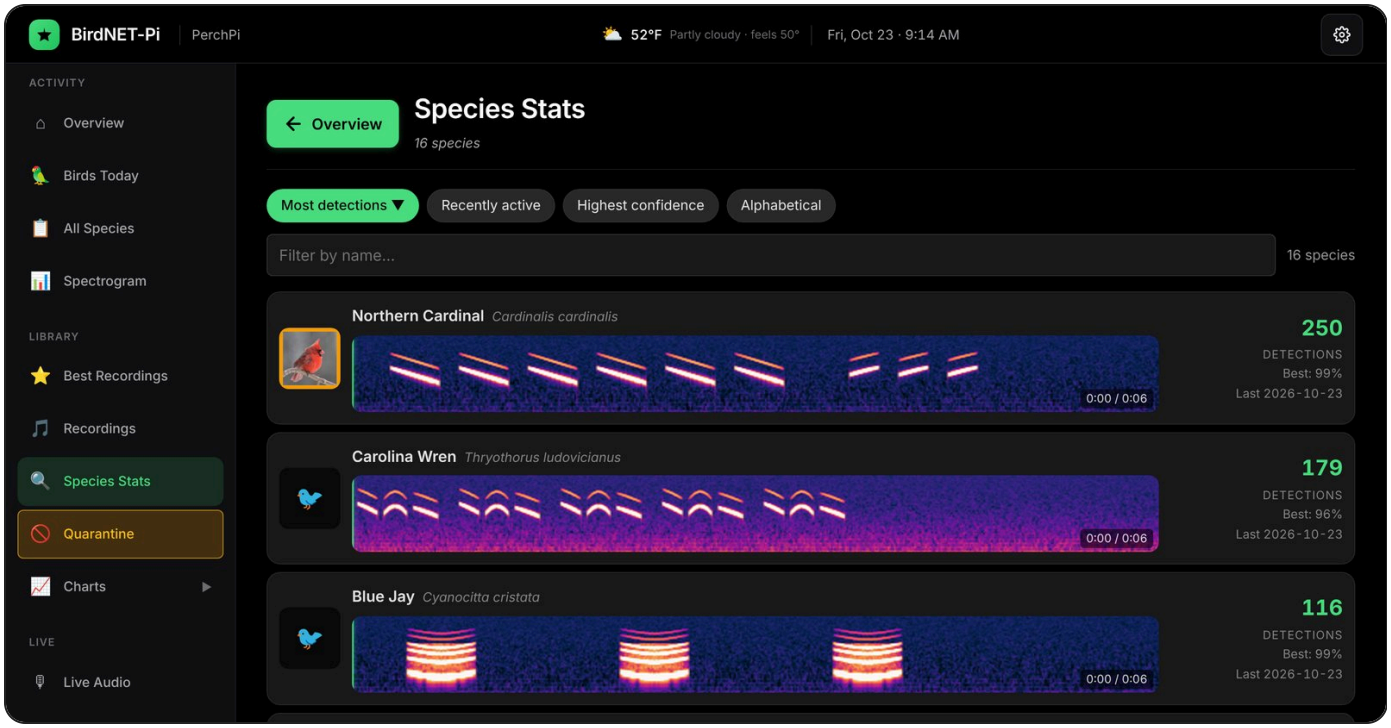
Birds and recordings



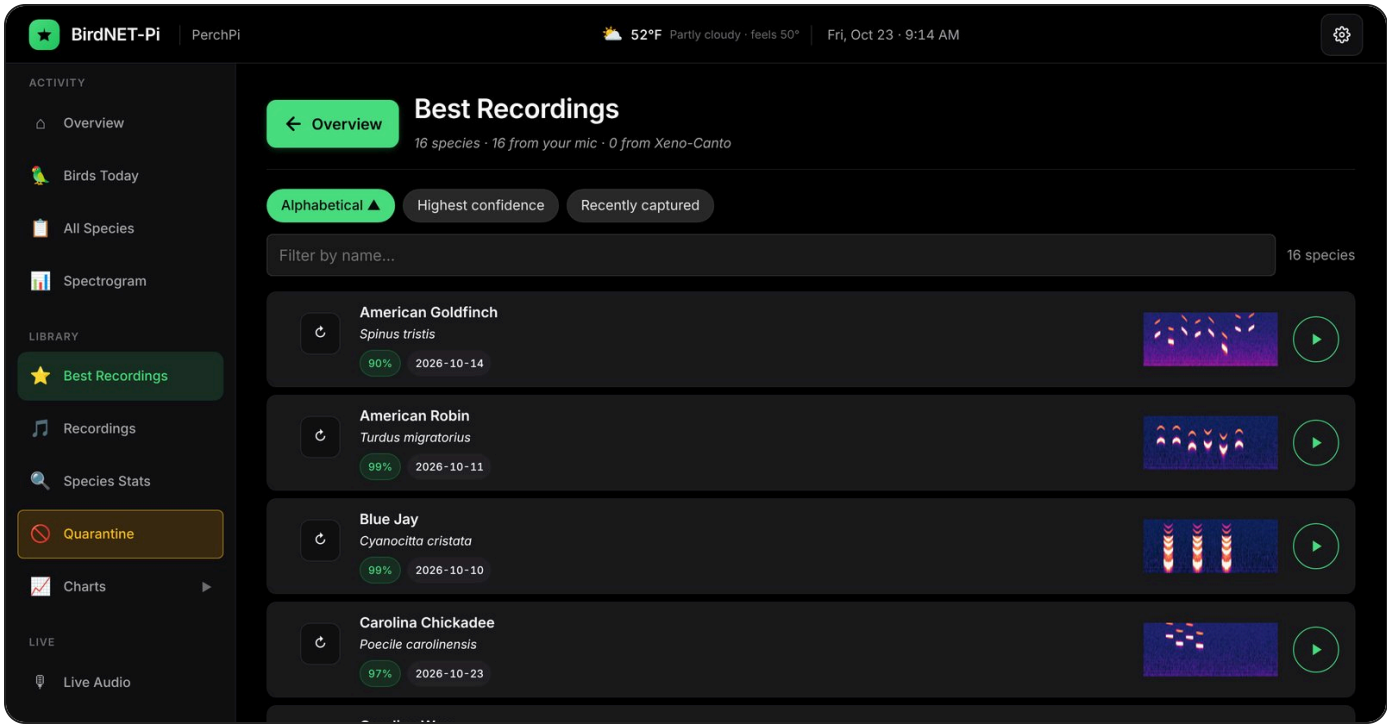
Birds Today: every species heard today. Sort by most detections, recently active, confidence, or A–Z. Tap ▶ to listen.



A bird's page: tap any bird for its detections by time of day (dawn, morning, afternoon, evening, night).



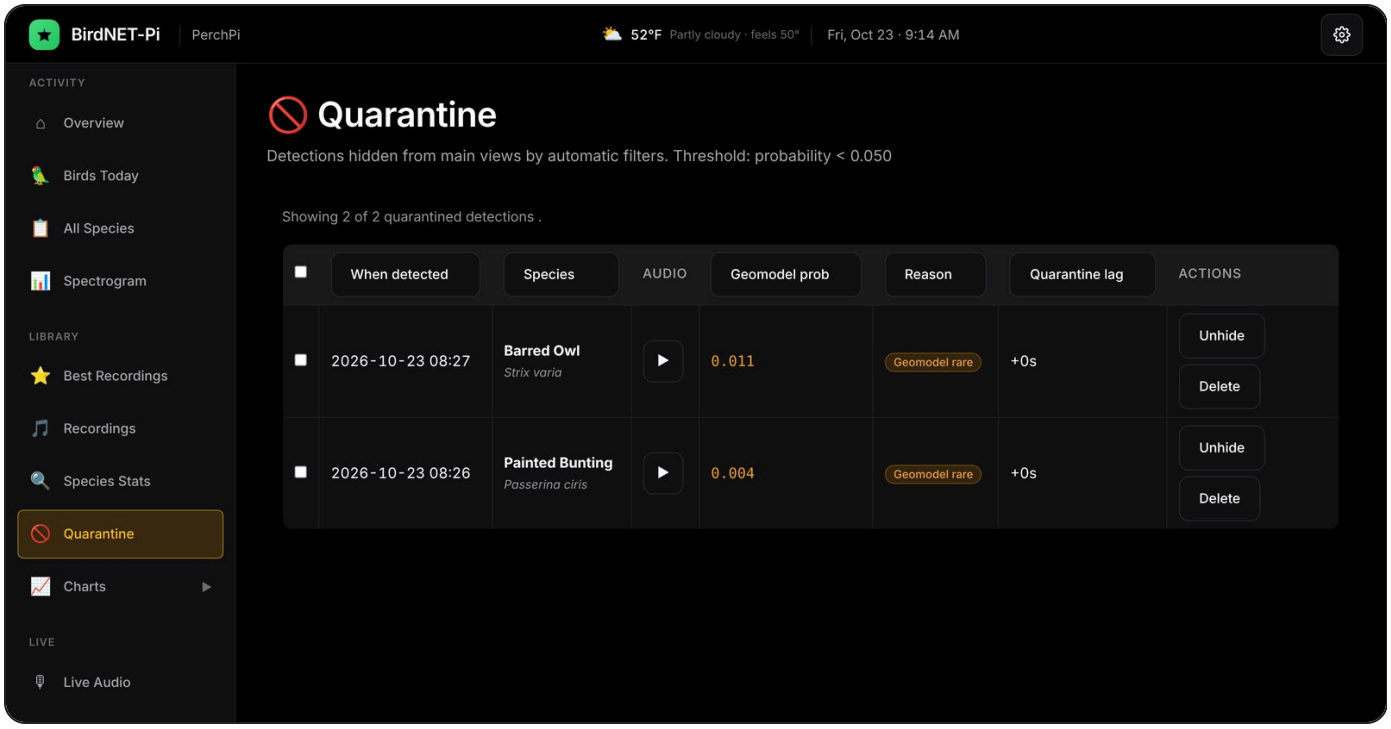
Species Stats: each species with its best recording's spectrogram, totals, and when it was last heard. The bright marks are the bird's song.



Best Recordings: the clearest recording of every species.

Reviewing Quarantine

PerchPi quarantines detections of birds that are very unlikely at your location this week. Most are mistakes (a similar-sounding bird, a noise); some are real rarities. When there's something to review, **Quarantine** turns orange in the menu.



Two detections waiting for review (example data).

1. Open **Quarantine** and press ▶ to play the recording.
2. **Unhide** if it really is that bird: it moves into your normal lists.
3. **Delete** if it's wrong. This can't be undone.

> Quarantine works well only once your **location is set** .

Touchscreen and phone

With a screen attached, PerchPi shows the Overview full-screen from the moment it starts, like the cover of this guide. Tap a bird to see it and play its song. The touchscreen is for watching and listening; use a phone or computer to change settings.

> **Tip:** on a phone, add the PerchPi page to your home screen so it opens like an app.

Settings you might want

All under the gear icon. Scroll to the section, change it, and click **Update Settings**.

SETTING	WHERE	WHAT IT DOES
Location	Settings → Location	Latitude and longitude (4 decimals). Set this first: it improves accuracy a lot.
Site Name	Settings → Location	The name in the browser tab and header.
Web password	Advanced Settings → BirdNET-Pi Password	Letters and numbers only. Clear the box first.
BirdWeather	Settings → BirdWeather	Share detections on the BirdWeather community map. Needs a free station token from birdweather.com.
Notifications	Settings → Notifications	Messages for new species, the first detection of each species each day, or every detection, by email, Telegram, Pushover, ntfy, and more. **Send Test Notification** checks it works.
Bird-name language	Settings → Localization	The language of bird names.
Time zone	Settings → Time and Date	Should match what you chose in Imager.
Audio kept	Advanced Settings → PerchPi Retention	How many days of audio to keep (default 45).

> Leave **Minimum Confidence** , **Sigmoid Sensitivity**, and **Overlap** in Advanced Settings at their defaults unless you know why you're changing them. They're tuned for the recommended microphone.

Storage: what's kept

- The **list of detections** is kept indefinitely.
- **Audio clips** are kept for **45 days**, but PerchPi always keeps the **5 best recordings of every species**.
- If the card gets nearly full, PerchPi deletes the oldest unprotected clips early, keeping at least 15% free.
- **gear → System Info** shows how full the card is. 64 GB or more gives plenty of room.

Turning PerchPi off and on



gear → System Controls.

> **Never just pull the plug** while it's running: that can damage the memory card. And **never press Clear ALL data**: it erases every detection and recording.

- **Turn off:** `gear → System Controls → Shutdown`. Wait about 30 seconds, until the Pi's light stops blinking green, then unplug.
- **Turn on:** plug the power back in.
- **Restart:** `System Controls → Reboot`.
- After a power cut, PerchPi starts again by itself.

Backing up and updating

`gear → System Controls → Backup data` downloads one file with your **detection database, recordings, notification setup, and species lists**. It can be several gigabytes; use a computer.

> **In v1, write down your settings too.** The backup does not include the Settings and Advanced Settings choices (location, web password, site name, BirdWeather token, retention days). And **Restore data** has a known problem in v1: please email before using it.

Updating: v1 has no update button. A new version is written onto the card with Imager, which replaces everything on it; its release notes will explain how to keep your data. New versions: github.com/churchja/perchpi-release/releases

Troubleshooting

WHAT YOU SEE	WHAT TO DO
The page won't open	Check power and that you're on the same Wi-Fi. Wait 5 minutes after a restart. If `.local` doesn't work on your device, use the Pi's number address from your router, e.g. `http://192.168.1.57/`.
It keeps asking for a password	Use the **web** login (`birdnet` and your web password), not the one you typed in Imager.
No new birds	Open **Live Audio**. Silence = microphone problem: check the **red** jack, then gear → System Controls → Reboot. Sound but no birds: check your location, and wait for a busy morning (birds are most active after sunrise).
Many wrong birds	Set your location. Move the microphone away from noise. Review Quarantine.
A bird you heard is missing	Look in **Quarantine**.
The screen is black	Give it 5 minutes. Plug the screen in before powering the Pi, then Reboot.
Imager doesn't show "PerchPi v1"	Repeat step 4: whole address pasted, then **Apply & Restart**.
Imager didn't ask for hostname and Wi-Fi	You probably used "Use custom". Repeat steps 4 and 5.
Forgot the web password	Email for help (below).
The card is full	gear → System Info. Lower the retention days in Advanced Settings, or use a bigger card.

Getting help

Email **perchpi@shadyacres.my**. Include what you were doing and what happened, a photo or screenshot of the screen, and if you can, a copy of **View Log**. PerchPi is a hobby project maintained by one person, so allow a few days for a reply.

PerchPi download page
github.com/churchja/perchpi-release
Guides, release notes, hood print files.

About this guide

Screens show **example data**, made with PerchPi v1's own pages; your station will show your own birds. Spectrograms in the examples are illustrative. Northern Cardinal photo: U.S. Geological Survey, Columbia Environmental Research Center (public domain). PerchPi builds on BirdNET-Pi and BirdNET (Cornell Lab of Ornithology, K. Lisa Yang Center for Conservation Bioacoustics) and uses Google's Perch model. Free for personal, non-commercial use under CC BY-NC-SA 4.0; see LICENSE and NOTICE on the download page.